

**Question no 1**

Implement CIDR (Classless Inter-Domain Routing) allocation for IP range 10.0.0.0/8 to create subnet for a company with 62 host requirements for a network. Provide subnet table.

- What will be Custom Subnet Mask?
- What will be Subnet ID?
- What will be Broadcast Address?
- What is Total Host Range and Valid/Usable Host Range?

**Answer:****(a) What will be the Custom Subnet Mask?**

- To accommodate 62 hosts:  
 $2^n - 2 \geq 62 \Rightarrow n = 6$
- Host bits = 6  $\Rightarrow$  Network bits =  $32 - 6 = 26$ .
- Therefore, the **custom subnet mask is 255.255.255.192 (/26)**.

**(b) What will be Subnet ID?**

- The first subnet ID will be: **10.0.0.0/26**
- Block size = 64 in the last octet.

**(c) What will be Broadcast Address?**

- For subnet 10.0.0.0/26, the broadcast address is **10.0.0.63**.

**(d) What is Total Host Range and Valid/Usable Host Range?**

- Total address range: **10.0.0.0 – 10.0.0.63**
- Usable range: **10.0.0.1 – 10.0.0.62**
- Number of usable hosts: **62**

**Step 5: Subnet Table (First 4 Subnets)**

| Subnet ID     | Broadcast  | Valid Host Range        | Subnet Mask     | Usable Hosts |
|---------------|------------|-------------------------|-----------------|--------------|
| 10.0.0.0/26   | 10.0.0.63  | 10.0.0.1 – 10.0.0.62    | 255.255.255.192 | 62           |
| 10.0.0.64/26  | 10.0.0.127 | 10.0.0.65 – 10.0.0.126  | 255.255.255.192 | 62           |
| 10.0.0.128/26 | 10.0.0.191 | 10.0.0.129 – 10.0.0.190 | 255.255.255.192 | 62           |
| 10.0.0.192/26 | 10.0.0.255 | 10.0.0.193 – 10.0.0.254 | 255.255.255.192 | 62           |

**Question no 2**

Your organization is expanding to a new floor and requires seamless network connectivity.

Currently, all employees are connected using a single switch and a Class C IP address range. You are given a network **192.168.100.0/24**. Design a subnetting plan for the following departments:

| Department | No. of Hosts |
|------------|--------------|
| Admin      | 50           |
| Sales      | 15           |
| IT         | 25           |

- Subnet the given IP block using **VLSM** to allocate addresses efficiently.
- Mention the **subnet ID**, **broadcast address**, **range of usable IPs**, and **subnet mask** for each department
- Justify why **VLSM** is more efficient than fixed-size subnetting in this case.

**Answer:****Step 1: Arrange Requirements in Descending Order**

- **Admin:** 50 hosts
- **IT:** 25 hosts
- **Sales:** 15 hosts

**(a)****Step 2: Allocate Subnets Using VLSM**

- Admin (50 hosts):**
  - Required: 50
  - Closest subnet:  $2^6 - 2 = 62$  hosts → **/26**
  - Subnet ID: 192.168.100.0/26
  - Range: 192.168.100.1 – 192.168.100.62
  - Broadcast: 192.168.100.63
- IT (25 hosts):**
  - Required: 25
  - Closest subnet:  $2^5 - 2 = 30$  hosts → **/27**
  - Subnet ID: 192.168.100.64/27
  - Range: 192.168.100.65 – 192.168.100.94
  - Broadcast: 192.168.100.95
- Sales (15 hosts):**
  - Required: 15
  - Closest subnet:  $2^5 - 2 = 30$  hosts → **/27**
  - Subnet ID: 192.168.100.96/27
  - Range: 192.168.100.97 – 192.168.100.126
  - Broadcast: 192.168.100.127

**(b)****Step 3: Subnet Table**

| Department | Subnet ID        | Broadcast Address | Valid Host Range               | Subnet Mask     | Usable Hosts |
|------------|------------------|-------------------|--------------------------------|-----------------|--------------|
| Admin      | 192.168.100.0/26 | 192.168.100.63    | 192.168.100.1 – 192.168.100.62 | 255.255.255.192 | 62           |

|       |                   |                 |                                     |                 |    |
|-------|-------------------|-----------------|-------------------------------------|-----------------|----|
| IT    | 192.168.100.64/27 | 192.168.100.95  | 192.168.100.65 –<br>192.168.100.94  | 255.255.255.224 | 30 |
| Sales | 192.168.100.96/27 | 192.168.100.127 | 192.168.100.97 –<br>192.168.100.126 | 255.255.255.224 | 30 |

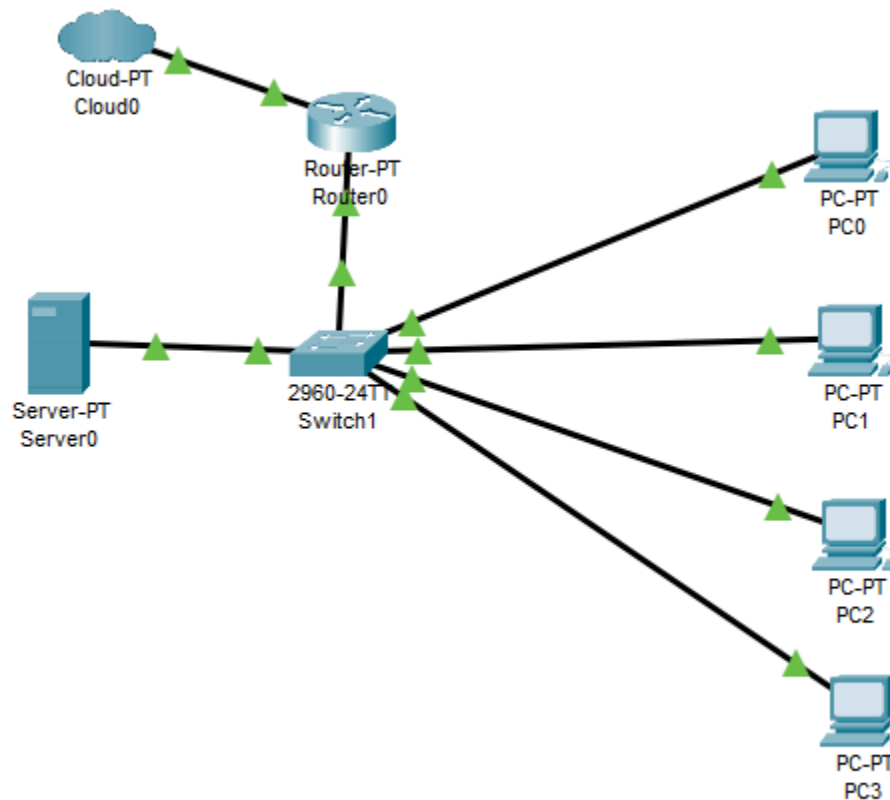
(c)

**Step 4: Justification – Why VLSM is Efficient**

If fixed-size subnetting were used, each department would be allocated a /26, wasting many IP addresses (e.g., Sales requires only 15 but gets 62). **VLSM** allocates variable-sized subnets based on actual need. This leads to efficient use of the 192.168.100.0/24 block. **VLSM** improves address utilization, reduces wastage, and keeps routing tables more optimized.

**Question#3:**

Create a mini project: Design and simulate a small office network with DHCP, DNS, and NAT enabled. Use Packet Tracer or similar tools.

**Answer:**

To design and simulate a small office network with DHCP, DNS, and NAT, the following steps were performed in Cisco Packet Tracer:

**Step 1: Network Topology**

- Devices used: 1 Router-PT, 1 Switch-2960, 4 PCs, 1 Server, and 1 Cloud (Internet).
- PCs and Server connected to Switch.
- Switch connected to Router Fa0/0.

- Router Fa1/0 connected to Cloud.

### Step 2: DHCP and DNS Server Configuration

- DHCP Pool Name: *office*
- Default Gateway: 192.168.1.1
- DNS Server: 192.168.1.2
- Starting IP Address: 192.168.1.100
- Subnet Mask: 255.255.255.0
- Maximum Users: 50
- Enabled DNS, HTTP, and HTTPS services.

### Step 3: Router Configuration (NAT Setup)

On the router CLI:

```
Router> enable
```

```
Router# configure terminal
```

#### ! Assign IP addresses

```
Router(config)# interface fastEthernet0/0
```

```
Router(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

```
Router(config)# interface fastEthernet1/0
```

```
Router(config-if)# ip address 200.1.1.1 255.255.255.0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

#### ! Define inside/outside

```
Router(config)# interface fastEthernet0/0
```

```
Router(config-if)# ip nat inside
```

```
Router(config-if)# exit
```

```
Router(config)# interface fastEthernet1/0
```

```
Router(config-if)# ip nat outside
```

```
Router(config-if)# exit
```

#### ! NAT Configuration

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
```

```
Router(config)# ip nat inside source list 1 interface fastEthernet1/0  
overload
```

### Step 4: Testing

- PCs automatically received IPs via DHCP.
  - Successful ping to Gateway (192.168.1.1) and Server (192.168.1.2).
  - DNS resolved domain names (e.g., *www.office.com*).
  - HTTP/HTTPS access to the server was successful.
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